

- 10 (a)** e.g. large bandwidth/carries more information
 low attenuation of signal
 low cost
 smaller diameter, easier handling, easier storage, less weight
 high security/no crosstalk
 low noise/no EM interference
(allow any four sensible suggestions, 1 each, max 4) B4 [4]
- (b) (i)** infra-red B1 [1]
- (ii)** lower attenuation than for visible light B1 [1]
- (c) (i)** gain/dB = $10 \lg(P_2/P_1)$ C1
 $26 = 10 \lg(P_2/9.3 \times 10^{-6})$
 $P_2 = 3.7 \times 10^{-3} \text{ W}$ A1 [2]
- (ii)** power loss along fibre = $30 \times 0.2 = 6.0 \text{ dB}$ C1
either $6 = 10 \lg(P/3.7 \times 10^{-3})$ *or* $6 \text{ dB} = 4 \times 3.7 \times 10^{-3}$
or $32 = 10 \lg(P/9.3 \times 10^{-6})$
 input power = $1.5 \times 10^{-2} \text{ W}$ A1 [2]

Page 6	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – October/November 2011	9702	42